

Md Mahfuj Ahmed

Full Stack Developer · Web · AI Integration · Mobile · C++ / Embedded

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SUMMARY

Full-stack developer with professional and project experience shipping production applications end-to-end, from API and database design to frontend delivery. Works across web (TypeScript, Vue, React, Next.js, Node.js, Django), AI integration (Google Gemini, Whisper speech-to-text), mobile app development, and C++/embedded systems (Arduino, I2C sensors). Comfortable with CI/CD, automated testing, and cloud deployment. Diploma in Full Stack Web Development from Red River College Polytechnic; Group Champion, Sprint to Innovation 2025. Builds AI-first, using Claude Code in a daily development workflow.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, C++17, SQL

Frontend: Vue.js, React, Next.js, Tailwind CSS, Pinia, TanStack Query

Backend & APIs: Node.js, Netlify Functions (serverless), Django, Django REST Framework, REST API design, GraphQL

Databases & Auth: PostgreSQL, MySQL, MongoDB, Firebase, Supabase, Prisma, NextAuth

AI Integration: Google Gemini, Groq / Whisper speech-to-text, prompt and API-contract design

Mobile: Mobile application development

Embedded & Systems: C++17, CMake, Arduino, I2C sensor integration (BME280), unit testing

Tools & DevOps: Git, GitHub (PRs, code review), GitHub Actions CI/CD, Playwright, Docker, Stripe API, Vite, Vercel, Netlify, AWS

PROJECT EXPERIENCE

EventFlow — Full Stack · Vue 3 · TypeScript · Firebase · MongoDB · Netlify Functions · Stripe

[GitHub](#) · [Live](#)

- Built a full-stack event ticketing and CRM platform end-to-end: Vue 3/TypeScript frontend, TypeScript serverless backend on Netlify Functions, MongoDB for event and order data, and Firebase authentication.
- Implemented guest checkout with Stripe, an email-matched account-claim flow, and a host onboarding and approval pipeline with automated notification emails.
- Shipped a host dashboard with live ticket-sales and revenue reporting, plus full event CRUD with sold-ticket safeguards.
- Built a 10-spec Playwright E2E suite gated by GitHub Actions CI on every pull request against a dedicated test database; merged through 8 reviewed PRs.

RoleReady — AI Platform · Next.js · TypeScript · Prisma/Neon · NextAuth v5 · Google Gemini · Groq Whisper

[GitHub](#) · [Live](#)

- Designed and built the complete backend for an AI interview-coaching platform: API routes for resume and job-description intake, gap analysis, and tailored question generation on Prisma/Neon (PostgreSQL) with NextAuth v5.
- Integrated Groq Whisper for real-time speech-to-text and Google Gemini for structured feedback scoring (Relevance, Depth, Communication), defining the API contract between frontend and AI services.
- Shipped session reporting (overall score, per-category breakdown, improvement insights) with an accessible UI, deployed on Vercel.

AV Device Controller — C++ / Embedded · C++17 · CMake · Arduino · I2C · BME280

[GitHub](#)

- Built a C++17 control system for multiple venue AV displays (power, volume, brightness, input switching) with per-device and broadcast commands and a config-driven device list editable without recompiling.
- Designed a layered architecture (parser → controller → device, with logging and alert modules) so the CLI can be swapped for a web UI or REST API without changing control logic; no external dependencies.
- Extended temperature monitoring to real hardware: an Arduino Uno reads a BME280 sensor over I2C and triggers the same threshold alerts as the simulation.
- Added timestamped audit logging of confirmed state changes, layered input validation, parser unit tests, and file-driven integration tests.

PROFESSIONAL EXPERIENCE

Junior Software Engineer · LILAB · Sylhet, Bangladesh

Feb 2021 – Oct 2022

- Developed RESTful APIs with Django for a high-traffic e-commerce platform, enabling efficient data synchronization between backend services and frontend clients.
- Built full-stack e-commerce features with Next.js, TypeScript, and React Query, pairing scalable data fetching with backend API work.
- Automated CI/CD pipelines with GitHub Actions for zero-downtime deployments.
- Collaborated in code reviews, database schema design, and UX research within a cross-functional team.

EDUCATION

Diploma in Full Stack Web Development · Red River College Polytechnic, Winnipeg, MB

Sep 2023 – Aug 2025

- Group Champion — Sprint to Innovation 2025 (Canada Life, Tech Manitoba & RRC)
- AWS Academy Cloud Foundations Certification